

Exploratory and Statistical Data Analysis with Python

Exploratory Data Analysis, Correlation Analysis, and Statistical Analysis

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Course Modules

- ① Exploratory Data Analysis
- ② Exploring Correlation in Python
- ③ What Is Statistical Analysis?

Exploratory Data Analysis

Lesson Introduction

This lesson introduces **Exploratory Data Analysis (EDA)** as a foundational stage before statistical testing or machine-learning modeling. It focuses on examining data structure, identifying distributions, detecting missing data and outliers, assessing relationships among variables, transforming data, and communicating important findings.

The lesson combines conceptual knowledge with Python examples so that learners can carry out a complete EDA workflow.

Learning Outcomes

After completing this lesson, learners will be able to:

- ▶ Explain the purpose and role of EDA in the data analysis process.
- ▶ Distinguish among univariate, bivariate, and multivariate analysis.
- ▶ Check the size, data types, missing values, and initial characteristics of a DataFrame.
- ▶ Use descriptive statistics to assess central tendency, dispersion, skewness, and kurtosis.
- ▶ Select histograms, box plots, bar charts, scatter plots, heatmaps, and pair plots for different analytical purposes.
- ▶ Identify and handle missing data using context-appropriate methods.
- ▶ Detect outliers using the IQR rule, Z-scores, and box plots.
- ▶ Apply transformations such as min-max scaling, standardization, and one-hot encoding.
- ▶ Use Pandas, Matplotlib, Seaborn, Plotly, and Scikit-learn at a basic level for EDA.
- ▶ Interpret results carefully, especially by distinguishing correlation from causation.
- ▶ Present findings, limitations, and recommended next steps in an EDA report.

Lesson Structure

The lesson covers:

- 1 The concept and importance of EDA.
- 2 Univariate, bivariate, and multivariate analysis.
- 3 Common tools in Python.
- 4 Understanding the problem and the meaning of the data.
- 5 Importing and inspecting data.
- 6 Handling missing data.
- 7 Exploring distributions and statistical characteristics.
- 8 Transforming and encoding data.
- 9 Visualizing relationships among variables.

Lesson Structure

- ⑩ Detecting and handling outliers.
- ⑪ Communicating findings and insights.
- ⑫ Applications, review questions, and practical exercises.

Prerequisites

Learners should have:

- ▶ Basic Python knowledge.
- ▶ An introductory understanding of DataFrames and data types.
- ▶ Access to Jupyter Notebook, JupyterLab, or Google Colab.
- ▶ The pandas, matplotlib, seaborn, plotly, and scikit-learn libraries.

Install the required libraries with:

```
pip install pandas matplotlib seaborn plotly scikit-learn
```

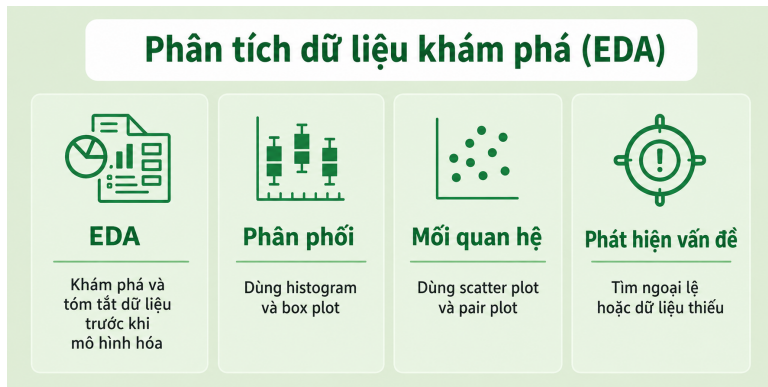
Exploratory Data Analysis, commonly called **EDA**, is an important stage in the data analysis process. EDA helps analysts explore, summarize, and visualize data in order to:

- ▶ Understand the structure of the dataset.
- ▶ Detect patterns and trends.

Prerequisites

- ▶ Identify unusual values.
- ▶ Check initial assumptions.
- ▶ Evaluate relationships among variables.
- ▶ Prepare data before applying statistical or machine-learning models.

Exploratory data analysis overview



Exploratory data analysis overview

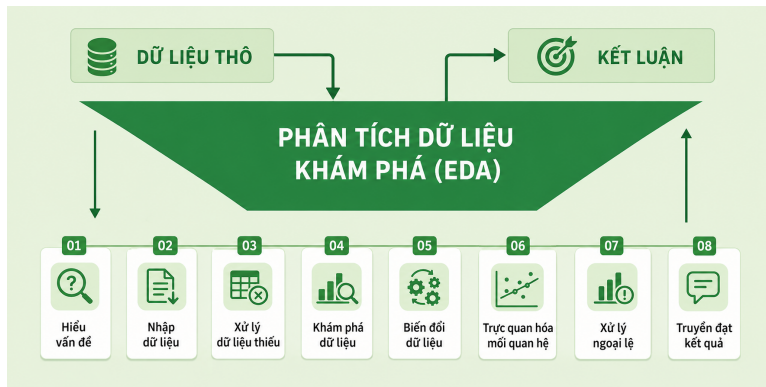
Quick Check - Prerequisites

Question 1. What is the main objective of EDA?

- A. Only to build machine-learning models
- B. To explore and understand data before modeling
- C. Only to store data
- D. To completely replace data cleaning

Question 2. True or false? EDA is usually performed before statistical or machine-learning modeling.

Importance of EDA



Importance of EDA

The Importance of EDA and Main Benefits

EDA is important because it helps analysts understand a dataset before drawing conclusions or building models. It gives the analyst an early view of structure, quality, patterns, and modeling implications.

- ▶ Provides a clear view of the number of variables, data types, and data distributions.
- ▶ Detects patterns and relationships among variables.
- ▶ Identifies data errors and outliers that may affect results.
- ▶ Highlights potentially important features for modeling.
- ▶ Supports the selection of suitable modeling methods.

Example

A customer dataset may contain variables such as age, income, residential area, and spending level. EDA can help determine which customer groups spend the most, whether income is related to spending, whether unusually large income values exist, and whether some variables contain too much missing data.

Quick Check - Main Benefits

Question 1. How can EDA support model selection?

- A. By identifying the characteristics and structure of the data
- B. By deleting all data
- C. By replacing model evaluation
- D. By only increasing the number of variables

Question 2. Which of the following can be detected through EDA?

- A. Outliers
- B. Relationships among variables
- C. Inappropriate data types
- D. All of the above

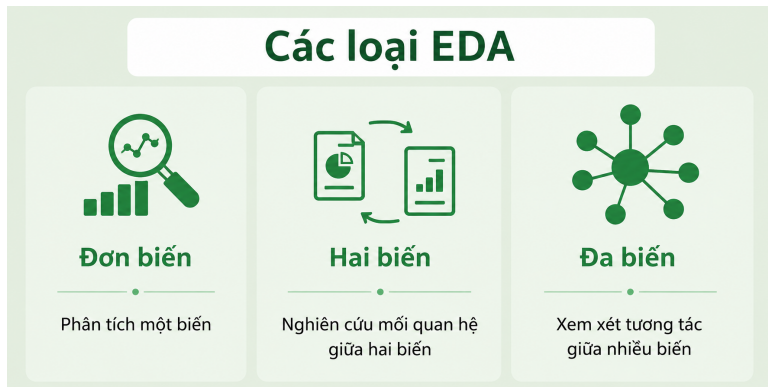
Question 3. Case. A column contains 70% missing values. How can EDA support a decision about this column?

Types of Exploratory Data Analysis

EDA is commonly divided into three types:

- 1 Univariate analysis.
- 2 Bivariate analysis.
- 3 Multivariate analysis.

Types of exploratory data analysis



Types of exploratory data analysis

1. Univariate Analysis

Univariate analysis examines one variable at a time to understand its characteristics and distribution.

Common Techniques

- ▶ **Histogram:** Shows the distribution of numerical values.
- ▶ **Box plot:** Shows dispersion and helps detect outliers.
- ▶ **Bar chart:** Commonly used for categorical variables.
- ▶ **Descriptive statistics:** Includes the mean, median, mode, standard deviation, and quantiles.

Example

For the variable age, an analyst can:

- ▶ Calculate the mean and median age.
- ▶ Draw a histogram to inspect the distribution.

1. Univariate Analysis

- ▶ Draw a box plot to detect unusual ages.
- ▶ Check whether the distribution is skewed.

Quick Check - 1. Univariate Analysis

Question 1. How many variables does univariate analysis examine at a time?

- A. One
- B. Two
- C. Three
- D. None

Question 2. Which chart is appropriate for examining the distribution of a numerical variable?

- A. Histogram
- B. Network diagram
- C. Heatmap
- D. Gantt chart

Question 3. What is a box plot commonly used for?

- A. Checking network connectivity

Quick Check - 1. Univariate Analysis

- B. Detecting outliers and examining dispersion
- C. Loading data
- D. Converting data types

2. Bivariate Analysis

Bivariate analysis examines the relationship between two variables to understand how they interact or change together.

Common Techniques

- ▶ **Scatter plot:** Shows the relationship between two numerical variables.
- ▶ **Correlation coefficient:** Measures the strength and direction of a linear relationship.
- ▶ **Cross-tabulation:** Shows the relationship between two categorical variables.
- ▶ **Line graph:** Compares two variables over time.
- ▶ **Covariance:** Indicates how two variables vary together.

Example

For age and income, an analyst can:

- ▶ Draw a scatter plot.
- ▶ Calculate the correlation coefficient.

2. Bivariate Analysis

- ▶ Check whether income tends to increase with age.
- ▶ Detect observations that differ clearly from the overall pattern.

Quick Check - 2. Bivariate Analysis

Question 1. Which technique is suitable for examining the relationship between two numerical variables?

- A. Scatter plot
- B. One-variable bar chart
- C. One-variable histogram
- D. Single frequency table

Question 2. A correlation coefficient measures:

- A. The number of data rows
- B. The strength and direction of a relationship between two variables
- C. The number of missing values
- D. File size

Question 3. True or false? A high correlation always proves that one variable causes the other.

3. Multivariate Analysis

Multivariate analysis examines three or more variables to understand complex relationships in a dataset.

Common Techniques

- ▶ **Pair plot:** Displays relationships among many pairs of variables.
- ▶ **Principal Component Analysis (PCA):** Reduces dimensionality while attempting to preserve important information.
- ▶ **Spatial analysis:** Examines geographic patterns using maps and location data.
- ▶ **Correlation matrix:** Shows relationships among many numerical variables.
- ▶ **Clustering and classification:** May reveal complex structure in data.

Example

In customer data, an analyst may examine:

- ▶ Age.
- ▶ Income.

3. Multivariate Analysis

- ▶ Purchase frequency.
- ▶ Total spending.

This can help identify groups of customers with similar characteristics.

Quick Check - 3. Multivariate Analysis

Question 1. Multivariate analysis typically examines:

- A. One variable
- B. Two variables
- C. Three or more variables
- D. No variables

Question 2. PCA is mainly used to:

- A. Increase the number of data rows
- B. Reduce data dimensionality
- C. Delete all numerical variables
- D. Convert charts into tables

Question 3. What does a pair plot do?

- A. Displays relationships among multiple pairs of variables

Quick Check - 3. Multivariate Analysis

- B. Displays only one value
- C. Processes text only
- D. Checks data-access permissions

Tools for EDA in Python

EDA can be performed with many tools. In this course, the practical focus is on Python-based workflows.

- ▶ **Pandas:** data handling, table inspection, cleaning, grouping, and transformation.
- ▶ **Matplotlib:** basic plotting and figure customization.
- ▶ **Seaborn:** statistical visualization such as histograms, box plots, scatter plots, heatmaps, and pair plots.
- ▶ **Plotly:** interactive visualization for exploratory dashboards and richer inspection.

Quick Check - Python Tools

Question 1. Which Python library is commonly used to manipulate tabular data?

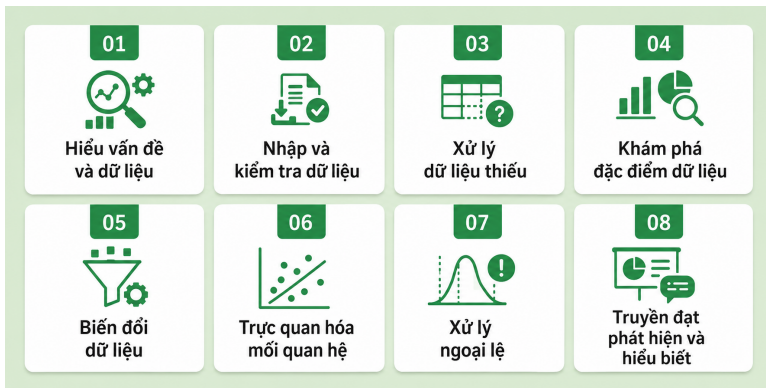
- A. Pandas
- B. Matplotlib
- C. Plotly
- D. Seaborn

Question 2. Which tool is suitable for interactive charts in Python?

- A. Plotly
- B. Pandas
- C. NumPy
- D. pathlib

EDA Workflow

EDA consists of a sequence of steps that help analysts understand data, detect problems, and prepare the data for subsequent analysis.



EDA workflow

Step 1. Understand the Problem and the Data

The first step is to understand the problem and the meaning of the available data.

Questions to Ask

- ▶ What is the objective or problem?
- ▶ Which variables are in the dataset?
- ▶ What does each variable represent?
- ▶ What types of data are present: numerical, categorical, text, or time?
- ▶ Are there quality problems or limitations?

Example

If the objective is to predict customer churn, identify:

- ▶ The target variable.

Step 1. Understand the Problem and the Data

- ▶ Potentially relevant input variables.
- ▶ The period during which the data were collected.
- ▶ Whether churners and non-churners are imbalanced.

Quick Check - Step 1. Understand the Problem and the Data

Question 1. Why is it necessary to understand the meaning of each variable?

- A. To interpret the data in the correct context
- B. To increase the number of variables
- C. To replace data collection
- D. To avoid visualization

Question 2. Which question is appropriate at this step?

- A. Which variable is the target?
- B. What data types are present?
- C. What limitations does the dataset have?
- D. All of the above

Step 2. Import and Inspect the Data

After understanding the problem, the data are loaded into Python or another analytical tool for an initial inspection.

Main Tasks

- ▶ Load the dataset correctly.
- ▶ Check the number of rows and columns.
- ▶ Identify missing values.
- ▶ Inspect each variable's data type.
- ▶ Detect errors, invalid values, or unusual observations.

Python Example

Step 2. Import and Inspect the Data

```
import pandas as pd

df = pd.read_csv("data.csv")

print(df.head())
print(df.shape)
print(df.info())
print(df.isnull().sum())
```

Explanation

- ▶ `head()` displays the first rows.
- ▶ `shape` returns the number of rows and columns.
- ▶ `info()` provides data-type information.
- ▶ `isnull().sum()` counts missing values in each column.

Quick Check - Step 2. Import and Inspect the Data

Question 1. Which attribute returns the number of rows and columns?

- A. `df.shape`
- B. `df.head()`
- C. `df.info()`
- D. `df.columns()`

Question 2. Which command helps inspect column data types?

- A. `df.info()`
- B. `df.plot()`
- C. `df.sort_values()`
- D. `df.drop()`

Question 3. Why should the data be inspected immediately after import?

Step 3. Handle Missing Data

Missing data are common in real datasets and can affect analytical quality.

Main Tasks

- ▶ Identify why values are missing.
- ▶ Decide whether to remove or impute missing values.
- ▶ Select an appropriate imputation method.
- ▶ Assess the uncertainty remaining after treatment.

Common Methods

- ▶ Mean imputation.
- ▶ Median imputation.
- ▶ Mode imputation.
- ▶ Regression-based imputation.
- ▶ K-nearest-neighbor imputation.

Step 3. Handle Missing Data

- ▶ Decision-tree methods.
- ▶ Removing rows or columns when appropriate.

Example

```
df["age"] = df["age"].fillna(  
    df["age"].median()  
)  
  
df["city"] = df["city"].fillna(  
    df["city"].mode()[0]  
)
```

Note

Removing missing observations may reduce sample size and introduce bias. Imputation may also change the original distribution. Therefore, the missingness mechanism and the analytical context should be considered.

Quick Check - Step 3. Handle Missing Data

Question 1. Why should analysts not automatically delete every row with missing data?

- A. It may reduce the sample and introduce bias
- B. Missing data are always correct
- C. Pandas cannot remove rows
- D. Every missing value equals zero

Question 2. Which method is suitable for a skewed numerical variable with outliers?

- A. Median
- B. Maximum value
- C. Random value
- D. Column name

Question 3. True or false? After imputation, all uncertainty is completely eliminated.

Step 4. Explore Data Characteristics

After handling missing data, examine the main statistical characteristics of the data.

Characteristics to Examine

- ▶ Data distribution.
- ▶ Mean, median, and mode.
- ▶ Standard deviation.
- ▶ Skewness.
- ▶ Kurtosis.
- ▶ Outliers and unusual observations.

Python Example

Step 4. Explore Data Characteristics

```
print(df.describe())  
print(df["income"].skew())  
print(df["income"].kurt())
```

Explanation

- ▶ `describe()` provides basic descriptive statistics.
- ▶ `skew()` measures distribution asymmetry.
- ▶ `kurt()` measures distribution kurtosis.

Quick Check - Step 4. Explore Data Characteristics

Question 1. Which measure describes data dispersion?

- A. Standard deviation
- B. Column name
- C. Number of rows
- D. File type

Question 2. Skewness measures:

- A. Distribution asymmetry
- B. Number of missing values
- C. Variable-name length
- D. Number of groups

Question 3. Why should both the mean and median be examined?

Step 5. Transform the Data

Data transformation makes the data more suitable for analysis or modeling.

Common Techniques

- ▶ Scaling.
- ▶ Min-max scaling.
- ▶ Standardization.
- ▶ One-hot encoding.
- ▶ Label encoding.
- ▶ Log transformation.
- ▶ Square-root transformation.
- ▶ Feature creation.
- ▶ Aggregation or grouping.

Min-Max Scaling Example

Step 5. Transform the Data

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df[["age", "income"]] = scaler.fit_transform(
    df[["age", "income"]]
)
```

One-Hot Encoding Example

```
df = pd.get_dummies(
    df,
    columns=["city"],
    drop_first=True
)
```

Feature Creation Example

Step 5. Transform the Data

```
df["revenue_per_order"] = (  
    df["total_revenue"] / df["number_of_orders"]  
)
```

Quick Check - Step 5. Transform the Data

Question 1. One-hot encoding is commonly used for:

- A. Categorical variables
- B. Image files
- C. Missing data
- D. Outliers

Question 2. Min-max scaling usually maps values to:

- A. 0 to 1
- B. 10 to 100
- C. Negative infinity to positive infinity
- D. Integers only

Question 3. Creating a new variable from existing variables is called:

- A. Feature engineering

Quick Check - Step 5. Transform the Data

- B. Data deletion
- C. File conversion
- D. Data duplication

Step 6. Visualize Data Relationships

Visualization helps reveal patterns and relationships that may be difficult to detect in tables.

Charts for Categorical Data

- ▶ Bar chart.
- ▶ Pie chart.
- ▶ Count plot.

Charts for Numerical Data

- ▶ Histogram.
- ▶ Box plot.
- ▶ Density plot.

Charts for Relationships

- ▶ Scatter plot.

Step 6. Visualize Data Relationships

- ▶ Line chart.
- ▶ Heatmap.
- ▶ Pair plot.

Python Examples

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.histplot(df["income"], kde=True)
plt.title("Income Distribution")
plt.show()
```

Step 6. Visualize Data Relationships

```
sns.scatterplot(  
    x="age",  
    y="income",  
    data=df  
)  
  
plt.title("Relationship Between Age and Income")  
plt.show()
```

```
sns.heatmap(  
    df.corr(numeric_only=True),  
    annot=True,  
    cmap="coolwarm"  
)  
  
plt.title("Correlation Matrix")  
plt.show()
```

Quick Check - Step 6. Visualize Data Relationships

Question 1. Which chart is suitable for examining the relationship between two numerical variables?

- A. Scatter plot
- B. Pie chart
- C. One-variable bar chart
- D. Frequency table

Question 2. A heatmap is often used to:

- A. Display a correlation matrix
- B. Delete missing values
- C. Load data from an API
- D. Convert data types

Question 3. Why can visualizations reveal problems that are not obvious in a numerical table?

Step 7. Handle Outliers

Outliers are observations that differ substantially from most of the data.

Possible Causes

- ▶ Data-entry errors.
- ▶ Measurement errors.
- ▶ Formatting errors.
- ▶ Rare real-world variation.
- ▶ Unusual but valid events.

Detection Methods

- ▶ Interquartile range (IQR).
- ▶ Z-score.
- ▶ Box plot.
- ▶ Domain-knowledge analysis.

Step 7. Handle Outliers

Treatment Methods

- ▶ Keep the value if it is valid.
- ▶ Correct it if it is a data-entry error.
- ▶ Apply capping.
- ▶ Transform the data.
- ▶ Remove it if it is clearly incorrect or harmful to the analysis.

IQR Example

Step 7. Handle Outliers

```
Q1 = df["income"].quantile(0.25)
Q3 = df["income"].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[
    (df["income"] < lower_bound) |
    (df["income"] > upper_bound)
]
```

Quick Check - Step 7. Handle Outliers

Question 1. Outliers may occur because of:

- A. Data-entry errors
- B. Real variation
- C. Measurement errors
- D. All of the above

Question 2. Which methods can detect outliers?

- A. IQR
- B. Z-score
- C. Box plot
- D. All of the above

Question 3. True or false? Every outlier must be removed.

Step 8. Communicate Results and Insights

The final EDA step is to present results clearly so that others can understand and use them.

What Should Be Reported?

- ▶ Analytical objective and scope.
- ▶ Problem context.
- ▶ Methods used.
- ▶ Main patterns and trends.
- ▶ Unusual observations.
- ▶ Data limitations.
- ▶ Recommendations for next steps.

Communication Principles

Step 8. Communicate Results and Insights

- ▶ Use appropriate charts.
- ▶ Avoid overcrowding a figure.
- ▶ Highlight important findings.
- ▶ Explain results in accessible language.
- ▶ State limitations and uncertainty clearly.

Quick Check - Step 8. Communicate Results and Insights

Question 1. What should appear in an EDA report?

- A. Analytical objective
- B. Main findings
- C. Data limitations
- D. All of the above

Question 2. Why should data limitations be stated?

- A. So readers understand the scope and reliability of the results
- B. To make the report longer
- C. To avoid presenting charts
- D. To eliminate analytical responsibility

Question 3. What type of language should a good EDA report use?

- A. Clear and understandable language

Quick Check - Step 8. Communicate Results and Insights

- B. Only complex terminology
- C. No explanation of charts
- D. Source code only

Applications of EDA

EDA is widely used across many fields because most analytical work begins with understanding the data before modeling or decision-making.

- ▶ **Market analysis and customer segmentation:** identify customer groups, purchasing behavior, and market trends.
- ▶ **Risk assessment in finance and insurance:** detect unusual transactions, risk factors, and high-risk customer groups.
- ▶ **Quality control in manufacturing:** identify product defects, process variation, and possible causes of quality problems.
- ▶ **Healthcare data analysis and disease prediction:** explore risk factors, disease trends, and relationships among symptoms, treatments, and outcomes.
- ▶ **Recommender systems and product optimization:** understand user behavior, engagement, and preferences.

Quick Check - Applications of EDA

Question 1. Detecting unusual transactions is a common EDA application in:

- A. Finance
- B. Graphic design
- C. Architecture
- D. Music

Question 2. In manufacturing, EDA can support:

- A. Product-defect detection
- B. Process-variation monitoring
- C. Quality control
- D. All of the above

Question 3. Case. An e-commerce platform wants to improve its product recommendation system. How can EDA help?

Content Summary

Topic	Main objective	Tool or technique
Univariate analysis	Understand one variable	Histogram, box plot, bar chart
Bivariate analysis	Understand relationships between two variables	Scatter plot, correlation, cross-tabulation
Multivariate analysis	Understand relationships among multiple variables	Pair plot, PCA, correlation matrix
Missing-data handling	Reduce the effect of incomplete data	Mean, median, mode, KNN
Feature exploration	Understand distribution and variability	Mean, median, standard deviation
Data transformation	Prepare data for analysis	Scaling, encoding, transformation
Outlier handling	Detect and assess unusual observations	IQR, Z-score, box plot
Communication	Convert results into insights	Charts, reports, dashboards

Exploring Correlation in Python

Lesson Introduction

This lesson introduces how to **explore correlation in Python** in order to evaluate the strength and direction of relationships between variables. It focuses on three widely used correlation coefficients-Pearson, Spearman, and Kendall-together with their implementation in Pandas, visualization with heatmaps, interpretation, limitations, and practical applications.

The lesson combines basic statistical concepts with Python examples so that learners can calculate, compare, and visualize correlation coefficients using a small dataset.

Learning Outcomes

After completing this lesson, learners will be able to:

- ▶ Explain positive correlation, negative correlation, and the absence of clear linear correlation.
- ▶ Interpret the range from -1 to 1 for a correlation coefficient.
- ▶ Distinguish among Pearson, Spearman, and Kendall correlation.
- ▶ Recognize when each method is appropriate.
- ▶ Calculate a correlation matrix with `DataFrame.corr()`.
- ▶ Calculate correlation between two columns with `Series.corr()`.
- ▶ Visualize a correlation matrix with `sns.heatmap()`.
- ▶ Interpret correlation values in context.
- ▶ Perform an initial check for multicollinearity.
- ▶ Recognize the effects of outliers and nonlinear relationships on Pearson correlation.
- ▶ Explain why correlation does not imply causation.
- ▶ Apply correlation analysis in feature selection, finance, healthcare, and recommender systems.

Lesson Structure

The lesson covers:

- 1 The concept and meaning of correlation.
- 2 Positive, negative, and no clear linear correlation.
- 3 Pearson correlation.
- 4 Spearman correlation.
- 5 Kendall correlation.
- 6 Comparison of the three methods.
- 7 Calculating correlation in Python.
- 8 Visualization with heatmaps.
- 9 Interpreting correlation values.

Lesson Structure

- 10 Detecting multicollinearity.
- 11 Limitations and applications.
- 12 Review questions and practical exercises.

Prerequisites

Learners should have:

- ▶ Basic Python knowledge.
- ▶ An introductory understanding of DataFrames and numerical variables.
- ▶ Basic knowledge of means, variance, and scatter plots.
- ▶ Access to Jupyter Notebook, JupyterLab, or Google Colab.
- ▶ The pandas, matplotlib, seaborn, scipy, and statsmodels libraries.

Install the required libraries with:

```
pip install pandas matplotlib seaborn scipy statsmodels
```

Correlation is one of the most widely used statistical measures for studying relationships between variables. In Python, correlation analysis helps determine whether two variables:

- ▶ Increase or decrease together.

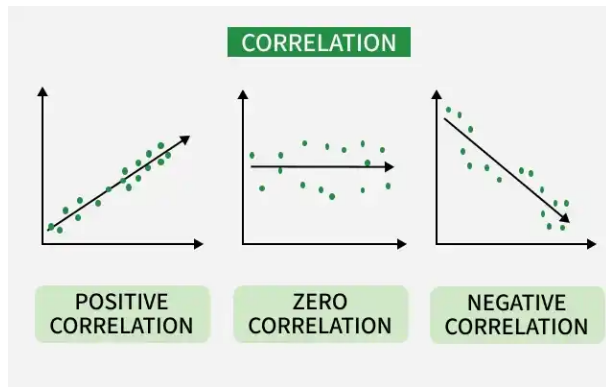
Prerequisites

- ▶ Move in opposite directions.
- ▶ Show no clear relationship.

Correlation analysis can support:

- ▶ Understanding relationships between variables.
- ▶ Feature selection in machine learning.
- ▶ Detection of multicollinearity.
- ▶ Data-driven decision-making.

Correlation overview



Correlation overview

Quick Check - Prerequisites

Question 1. What does correlation measure?

- A. The number of rows in a dataset
- B. The strength and direction of a relationship between two variables
- C. The size of a data file
- D. The number of missing values

Question 2. True or false? Correlation can support feature selection in machine learning.

What Is Correlation?

Correlation measures the **strength** and **direction** of the relationship between two numerical variables.

A correlation coefficient usually lies in the interval:

$$-1 \leq r \leq 1$$

where:

- ▶ $r = 1$: Perfect positive relationship.
- ▶ $r = -1$: Perfect negative relationship.
- ▶ $r = 0$: No clear linear correlation.

Positive Correlation

Positive correlation occurs when two variables tend to move in the same direction.

Examples:

- ▶ Height and weight.
- ▶ Study time and test scores.
- ▶ Advertising expenditure and sales in some contexts.

When one variable increases, the other tends to increase. When one decreases, the other tends to decrease.

Negative Correlation

Negative correlation occurs when two variables tend to move in opposite directions.

Examples:

- ▶ Price and demand.
- ▶ Speed and travel time over a fixed distance.
- ▶ Fuel efficiency and fuel consumption.

When one variable increases, the other tends to decrease.

No Clear Correlation

A correlation value near zero indicates no clear linear relationship.

Examples:

- ▶ Shoe size and examination score.
- ▶ Number of letters in a name and income.
- ▶ Favorite color and height.

Note

Note: A correlation coefficient of zero does not prove that two variables are completely unrelated. They may still have a nonlinear relationship.

Quick Check - No Clear Correlation

Question 1. A correlation of +1 represents:

- A. A perfect positive relationship
- B. A perfect negative relationship
- C. No correlation
- D. Missing data

Question 2. Which is an example of negative correlation?

- A. Height and weight
- B. Price and demand
- C. Study hours and test score
- D. Temperature and ice-cream sales

Question 3. True or false? A correlation coefficient of zero proves that two variables have no relationship of any kind.

Common Correlation Methods in Python

Python supports several methods for calculating correlation. Three common methods are:

- 1 Pearson correlation.
- 2 Spearman correlation.
- 3 Kendall correlation.

1. Pearson Correlation

Pearson correlation measures the **linear relationship** between two continuous variables.

Characteristics

- ▶ The value lies between -1 and +1.
- ▶ It is commonly used with continuous numerical data.
- ▶ It is suitable when the relationship is approximately linear.
- ▶ It is often associated with assumptions of approximately normal data.
- ▶ It is sensitive to outliers.

Formula

The Pearson correlation coefficient is:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

where:

1. Pearson Correlation

- ▶ x_i and y_i are observations.
- ▶ $\bar{x}$ and $\bar{y}$ are sample means.
- ▶ n is the number of observations.

When Should Pearson Be Used?

Pearson may be appropriate when:

- ▶ Both variables are numerical.
- ▶ The relationship of interest is linear.
- ▶ There are no severe outliers.
- ▶ The distributional conditions are suitable for the intended analysis.

Quick Check - 1. Pearson Correlation

Question 1. Pearson correlation primarily measures what type of relationship?

- A. Linear
- B. Categorical only
- C. Geographic
- D. Textual

Question 2. Pearson is most suitable for:

- A. Two continuous numerical variables
- B. Two text passages
- C. Two unordered nominal variables
- D. Two image files

Question 3. True or false? Pearson correlation may be strongly affected by outliers.

2. Spearman Correlation

Spearman correlation measures a **monotonic relationship** by converting values into ranks before calculating correlation.

Characteristics

- ▶ The relationship does not need to be linear.
- ▶ It is suitable for monotonic relationships.
- ▶ It can be used with ordinal data.
- ▶ It is useful when data are not normally distributed.
- ▶ It is often less sensitive to outliers than Pearson correlation.

Monotonic Relationship

A relationship is monotonic when:

- ▶ As one variable increases, the other consistently tends to increase; or
- ▶ As one variable increases, the other consistently tends to decrease.

2. Spearman Correlation

The rate of change does not have to be constant, and the relationship does not have to follow a straight line.

When Should Spearman Be Used?

Spearman may be appropriate when:

- ▶ The data are ordinal.
- ▶ The relationship is monotonic but nonlinear.
- ▶ The data do not satisfy normality assumptions well.
- ▶ Outliers substantially affect Pearson correlation.

Quick Check - 2. Spearman Correlation

Question 1. Spearman correlation is calculated from:

- A. Data ranks
- B. Column names
- C. Number of files
- D. Chart colors

Question 2. Spearman is suitable for:

- A. Monotonic relationships
- B. Only perfect linear relationships
- C. Only text data
- D. Only binary variables

Question 3. True or false? Spearman correlation can be used with ordinal data.

3. Kendall Correlation

Kendall correlation measures the consistency or agreement between ranked pairs of observations.

Characteristics

- ▶ It is rank-based.
- ▶ It measures agreement between pairs.
- ▶ It is often suitable for small datasets.
- ▶ It can be robust for ordinal data or data with ties.
- ▶ It may be slower on large datasets.

When Should Kendall Be Used?

Kendall may be considered when:

- ▶ The dataset is relatively small.

3. Kendall Correlation

- ▶ The data are ordinal.
- ▶ Rank consistency is of interest.
- ▶ A method with weaker distributional assumptions is preferred.

Quick Check - 3. Kendall Correlation

Question 1. Kendall correlation focuses on:

- A. Agreement between rankings
- B. The number of rows
- C. The mean
- D. File size

Question 2. Kendall is often suitable for:

- A. Small datasets
- B. Image data only
- C. Audio data only
- D. Data with no order

Question 3. What do Spearman and Kendall have in common?

- A. Both are rank-based

Quick Check - 3. Kendall Correlation

- B. Both apply only to text data
- C. Both only measure means
- D. Both always produce the same result as Pearson

Comparing Pearson, Spearman, and Kendall

Method	Relationship type	Suitable data	Key feature
Pearson	Linear	Continuous numerical variables	Popular and easy to interpret, but sensitive to outliers
Spearman	Monotonic	Numerical or ordinal data	Rank-based and suitable for monotonic nonlinear relationships
Kendall	Rank agreement	Ordinal data and small datasets	Robust and interpretable through concordant and discordant pairs

Quick Check - Comparing Pearson, Spearman, and Kendall

Question 1. Which method is most appropriate for a linear relationship between two continuous variables?

- A. Pearson
- B. Spearman
- C. Kendall
- D. None of the above

Question 2. Which method is appropriate for a monotonic but nonlinear relationship?

- A. Pearson
- B. Spearman
- C. Mean only
- D. Variance only

Question 3. Which method is often suitable for a small ordinal dataset?

- A. Kendall

Quick Check - Comparing Pearson, Spearman, and Kendall

- B. Pearson
- C. Histogram
- D. Min-max scaling

Calculating Correlation in Python: Sample Dataset

Python provides correlation functions in Pandas and visualization tools in Seaborn and Matplotlib. The example dataset contains scores in three subjects:

- ▶ Mathematics.
- ▶ Science.
- ▶ English.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    "Math": [78, 85, 96, 80, 86],
    "Science": [88, 90, 94, 82, 89],
    "English": [72, 75, 78, 70, 74]
}
```

Calculating Correlation in Python: Sample Dataset

```
df = pd.DataFrame(data)
```

```
df
```

Illustrative Result

Sample score dataset

	Math	Science	English
0	78	88	72
1	85	90	75
2	96	94	78
3	80	82	70
4	86	89	74

Sample score dataset

Quick Check - 1. Create a Sample Dataset

Question 1. Which object is used to create the table?

- A. `pd.DataFrame()`
- B. `plt.figure()`
- C. `sns.heatmap()`
- D. `df.drop()`

Question 2. How many variables are in the sample dataset?

- A. 2
- B. 3
- C. 5
- D. 15

Question 3. Why is this dataset suitable for demonstration?

2. Calculate Pearson Correlation

Pandas provides the `corr()` method to calculate correlations between numerical columns.

```
pearson_corr = df.corr(  
    method="pearson"  
)  
  
print(pearson_corr)
```

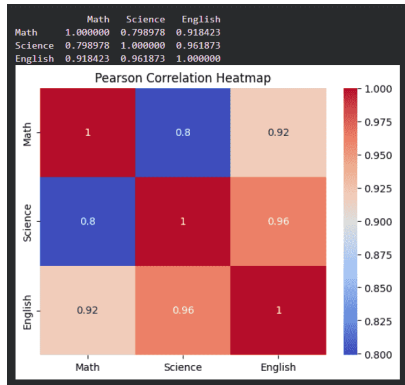
Visualize with a Heatmap

```
sns.heatmap(  
    pearson_corr,  
    annot=True,  
    cmap="coolwarm"  
)  
  
plt.title(  
    "Pearson Correlation Matrix"
```

2. Calculate Pearson Correlation

```
)  
plt.show()
```


Pearson correlation heatmap



Pearson correlation heatmap

2. Calculate Pearson Correlation

Explanation

- ▶ `df.corr(method="pearson")` calculates pairwise Pearson correlations.
- ▶ `annot=True` displays the correlation values in the cells.
- ▶ `cmap="coolwarm"` defines the color map.
- ▶ The main diagonal is always equal to 1 because each variable is perfectly correlated with itself.

Quick Check - 2. Calculate Pearson Correlation

Question 1. Which method calculates a correlation matrix?

- A. `df.corr()`
- B. `df.head()`
- C. `df.drop()`
- D. `df.merge()`

Question 2. Values on the main diagonal are usually:

- A. 0
- B. 1
- C. -1
- D. Undefined

Question 3. What does `annot=True` do?

- A. Displays values in the heatmap cells

Quick Check - 2. Calculate Pearson Correlation

- B. Removes missing values
- C. Converts data to strings
- D. Changes the number of rows

3. Calculate Spearman Correlation

Spearman converts values into ranks before calculating correlation.

```
spearman_corr = df.corr(  
    method="spearman"  
)  
  
print(spearman_corr)
```

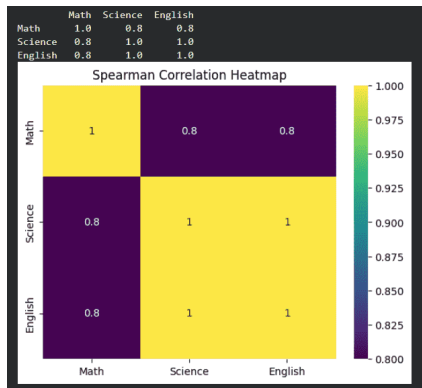
Visualization

```
sns.heatmap(  
    spearman_corr,  
    annot=True,  
    cmap="viridis"  
)  
  
plt.title(  
    "Spearman Correlation Matrix"
```

3. Calculate Spearman Correlation

```
)  
plt.show()
```

Spearman correlation heatmap



Spearman correlation heatmap

3. Calculate Spearman Correlation

Explanation

Spearman is appropriate when:

- ▶ Relationships are monotonic.
- ▶ Data are not normally distributed.
- ▶ Data are ordinal.
- ▶ Outliers substantially affect Pearson correlation.

Quick Check - 3. Calculate Spearman Correlation

Question 1. To calculate Spearman correlation in Pandas, method is set to:

- A. "spearman"
- B. "linear"
- C. "ranked"
- D. "ordinal"

Question 2. What does Spearman use before calculating correlation?

- A. Ranks
- B. Arithmetic means
- C. Variable names
- D. Row counts

4. Calculate Kendall Correlation

Kendall measures agreement between ranked observations.

```
kendall_corr = df.corr(  
    method="kendall"  
)  
  
print(kendall_corr)
```

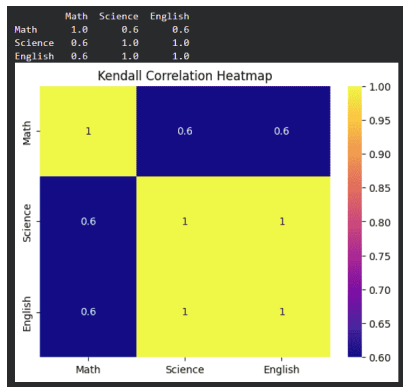
Visualization

```
sns.heatmap(  
    kendall_corr,  
    annot=True,  
    cmap="plasma"  
)  
  
plt.title(  
    "Kendall Correlation Matrix"
```

4. Calculate Kendall Correlation

```
)  
plt.show()
```

Kendall correlation heatmap



Kendall correlation heatmap

4. Calculate Kendall Correlation

Explanation

Kendall is useful when:

- ▶ The dataset is small.
- ▶ The data are ordinal.
- ▶ Rank agreement is important.
- ▶ Dependence on distributional assumptions should be reduced.

Quick Check - 4. Calculate Kendall Correlation

Question 1. Which value is used for method when calculating Kendall correlation?

- A. "kendall"
- B. "small"
- C. "pair"
- D. "ordinal"

Question 2. True or false? Kendall can be used to assess agreement between rankings.

5. Calculate Correlation Between Two Columns

Correlation between two specific columns can be calculated with `Series.corr()`.

```
corr_value = df["Math"].corr(  
    df["Science"]  
)  
  
print(  
    "Correlation between Math and Science:",  
    corr_value  
)
```

Create a Two-Column Matrix

5. Calculate Correlation Between Two Columns

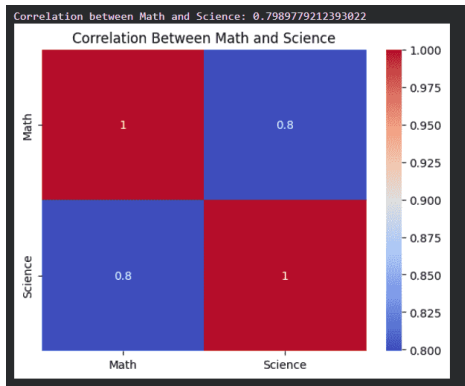
```
two_col_corr = df[
    ["Math", "Science"]
].corr()

sns.heatmap(
    two_col_corr,
    annot=True,
    cmap="coolwarm"
)

plt.title(
    "Correlation Between Math and Science"
)

plt.show()
```


Correlation between two columns



Correlation between two columns

5. Calculate Correlation Between Two Columns

Explanation

- ▶ `df["Math"].corr(df["Science"])` returns a single correlation value.
- ▶ `df[["Math", "Science"]].corr()` returns a 2 $\times$ 2 correlation matrix.
- ▶ A heatmap visually communicates the strength and direction of the relationship.

Quick Check - 5. Calculate Correlation Between Two Columns

Question 1. Which command directly returns the correlation between two columns?

- A. `df["Math"].corr(df["Science"])`
- B. `df["Math"].head(df["Science"])`
- C. `df.merge("Math", "Science")`
- D. `df.plot("Math", "Science")`

Question 2. A correlation matrix for two variables has size:

- A. 1×1
- B. 2×2
- C. 2×3
- D. 3×3

Interpreting Correlation Values

The following table provides a general interpretation.

Correlation value	Suggested interpretation
0.8 to 1.0	Strong positive correlation
0.5 to below 0.8	Moderate positive correlation
Above 0 to below 0.5	Weak positive correlation
0	No linear correlation
Above -0.5 to below 0	Weak negative correlation
Above -0.8 to -0.5	Moderate negative correlation
-1.0 to -0.8	Strong negative correlation

Note

Note: These thresholds are only guidelines. Interpretation depends on the field, sample size, data quality, and analytical objective.

Interpretation Examples

- ▶ 0.92: Strong positive correlation.
- ▶ 0.63: Moderate positive correlation.
- ▶ 0.18: Weak positive correlation.
- ▶ -0.74: Moderate negative correlation.
- ▶ -0.91: Strong negative correlation.
- ▶ 0.02: Almost no linear correlation.

Quick Check - Interpretation Examples

Question 1. A value of 0.87 is usually interpreted as:

- A. Strong positive correlation
- B. Strong negative correlation
- C. No correlation
- D. Weak negative correlation

Question 2. A value of -0.65 is usually interpreted as:

- A. Moderate negative correlation
- B. Moderate positive correlation
- C. Strong positive correlation
- D. No correlation

Question 3. Why should interpretation thresholds not be applied rigidly?

Visualizing Correlation with Heatmaps

A heatmap is commonly used to display a correlation matrix. It helps compare many pairwise relationships in one compact view.

```
corr_matrix = df.corr(  
    method="pearson"  
)  
  
sns.heatmap(  
    corr_matrix,  
    annot=True,  
    cmap="coolwarm",  
    vmin=-1,  
    vmax=1  
)  
  
plt.title(  
    "Correlation Heatmap"  
)
```

Visualizing Correlation with Heatmaps

```
plt.show()
```

Useful Parameters

- ▶ `annot=True`: Displays values inside the cells.
- ▶ `cmap`: Selects the color map.
- ▶ `vmin=-1`: Sets the minimum color-scale value.
- ▶ `vmax=1`: Sets the maximum color-scale value.
- ▶ `fmt=".2f"`: Displays two decimal places.
- ▶ `square=True`: Displays square cells.

More Complete Example

Visualizing Correlation with Heatmaps

```
sns.heatmap(  
    corr_matrix,  
    annot=True,  
    fmt=".2f",  
    cmap="coolwarm",  
    vmin=-1,  
    vmax=1,  
    square=True  
)  
  
plt.title(  
    "Correlation Matrix"  
)  
  
plt.show()
```

Quick Check - Heatmap Example

Question 1. Why should `vmin=-1` and `vmax=1` be used?

- A. To represent the full range of correlation values on the color scale
- B. To delete negative data
- C. To retain only strong values
- D. To rename columns

Question 2. What does `fmt=".2f"` do?

- A. Displays two decimal places
- B. Displays integers only
- C. Removes negative values
- D. Changes the data size

Detecting Multicollinearity

Multicollinearity occurs when two or more input variables are strongly correlated. It can make it difficult to isolate the effect of each variable and may make regression coefficients unstable.

Possible effects

- ▶ Model interpretation may become unreliable.
- ▶ Some variables may provide redundant information.
- ▶ Feature selection may be needed to simplify the model.

Initial check

A correlation matrix can be used to identify variable pairs with high absolute correlation.

```
corr_matrix = df.corr(numeric_only=True)
high_corr = corr_matrix.abs() > 0.8
print(high_corr)
```

Detecting Multicollinearity

Note

A correlation matrix is only an initial screening tool. A fuller multicollinearity assessment may require additional measures such as the Variance Inflation Factor.

Quick Check - Detecting Multicollinearity

Question 1. Multicollinearity occurs when:

- A. Input variables are strongly correlated with one another
- B. The dataset contains no numerical columns
- C. The data contain no missing values
- D. All variables are completely independent

Question 2. One possible effect of multicollinearity is:

- A. Model coefficients may become unstable
- B. Data automatically become more accurate
- C. Feature selection becomes unnecessary
- D. Every model achieves 100% accuracy

Limitations of Correlation Analysis

Correlation is useful for exploring associations, but it must be interpreted carefully.

- ▶ **Association only:** correlation indicates that two variables are related, but it does not prove causation. Ice-cream sales and fires may both rise in summer because high temperature affects both variables.
- ▶ **Sensitivity to outliers:** a few unusual observations may substantially change the Pearson coefficient.
- ▶ **Linear focus:** Pearson may be near zero even when two variables have a strong nonlinear relationship.
- ▶ **Dependence on data quality:** correlation may change with sample size, restricted ranges, errors, unrepresentative data, or omitted variables.
- ▶ **Practical importance:** a numerically strong correlation may still be unimportant without domain relevance.

Quick Check - Limitations of Correlation Analysis

Question 1. Why does correlation not prove causation?

- A. A third variable or a coincidental association may exist
- B. Correlation can never be calculated
- C. All variables are independent
- D. The coefficient is always zero

Question 2. Which type of relationship can Pearson fail to detect?

- A. A nonlinear relationship
- B. A linear relationship
- C. A perfect relationship
- D. A positive relationship

Question 3. True or false? A strong correlation always has major practical significance.

Feature Selection in Machine Learning

Correlation can help:

- ▶ Identify variables associated with a target.
- ▶ Detect redundant input variables.
- ▶ Reduce multicollinearity.
- ▶ Simplify a model.

Financial Market Analysis

Correlation is used to:

- ▶ Compare movements across assets.
- ▶ Support portfolio diversification.
- ▶ Evaluate relationships between indices and stocks.
- ▶ Analyze economic factors.

Correlation can support the study of relationships between:

- ▶ Age and disease risk.
- ▶ Drug dosage and response.
- ▶ Lifestyle and health indicators.
- ▶ Symptoms and treatment outcomes.

Recommender Systems

Correlation can be used to:

- ▶ Compare user behavior.
- ▶ Identify products rated similarly.
- ▶ Find groups of users with similar preferences.
- ▶ Support product or content recommendations.

Quick Check - Recommender Systems

Question 1. In machine learning, correlation can support:

- A. Feature selection
- B. Multicollinearity detection
- C. Identification of redundant variables
- D. All of the above

Question 2. In finance, correlation can be used to:

- A. Compare movements across assets
- B. Support portfolio diversification
- C. Analyze relationships across markets
- D. All of the above

Question 3. Case. A recommender system wants to identify users with similar preferences. How can correlation help?

Content Summary

Topic	Main meaning
Correlation	Measures the strength and direction of a relationship between two variables
Pearson	Measures linear relationships between numerical variables
Spearman	Measures monotonic relationships using ranks
Kendall	Measures agreement between rankings
Heatmap	Visualizes a correlation matrix
Multicollinearity	Input variables are strongly correlated with one another
Limitations	Correlation does not prove causation and may be affected by outliers
Applications	Machine learning, finance, healthcare, and recommender systems

What Is Statistical Analysis?

Lesson Introduction

This lesson introduces the foundations of **statistical analysis**, from collecting and organizing data to descriptive, inferential, exploratory, predictive, prescriptive, and causal methods. It emphasizes the use of data and evidence to support decisions, test hypotheses, evaluate uncertainty, and interpret results responsibly.

The lesson combines statistical concepts, basic formulas, Python examples, application scenarios, and quick-check questions. It provides learners with an overview of the role of statistics in data analysis, scientific research, and management.

Learning Outcomes

After completing this lesson, learners will be able to:

- ▶ Explain the concept and role of statistical analysis.
- ▶ Describe four major stages: data collection, data organization, data analysis, and interpretation and presentation.
- ▶ Distinguish between descriptive and inferential statistics.
- ▶ Distinguish between Exploratory Data Analysis (EDA) and Confirmatory Data Analysis (CDA).
- ▶ Recognize the roles of regression, hypothesis testing, confidence intervals, and ANOVA.
- ▶ Calculate and interpret the mean, variance, and standard deviation at a basic level.
- ▶ Distinguish predictive, prescriptive, and causal analysis.
- ▶ Explain the difference between correlation and causation.
- ▶ Recognize the basic uses of Python, SPSS, and Excel.
- ▶ Select suitable tools and methods for different analytical objectives.
- ▶ Present statistical results using reports, charts, dashboards, or presentations.

Learning Outcomes

- ▶ State limitations, uncertainty, and practical implications of results.
- ▶ Apply statistical concepts in business, healthcare, education, social science, and environmental contexts.

Lesson Structure

The lesson covers:

- 1 The concept and role of statistical analysis.
- 2 Data collection.
- 3 Data organization and cleaning.
- 4 Data analysis using EDA, CDA, regression, and hypothesis testing.
- 5 Interpretation and presentation.
- 6 Descriptive statistics.
- 7 Inferential statistics.
- 8 Exploratory Data Analysis.
- 9 Predictive modeling.

Lesson Structure

- 10 Prescriptive analysis.
- 11 Causal analysis.
- 12 Statistical analysis tools.
- 13 Importance and applications.
- 14 Review questions and case-based exercises.

Prerequisites

Learners should have:

- ▶ Basic mathematical knowledge of means and proportions.
- ▶ An introductory understanding of data, variables, samples, and populations.
- ▶ Basic Python knowledge for the coding examples.
- ▶ Access to Jupyter Notebook, JupyterLab, or Google Colab.
- ▶ The `numpy`, `pandas`, `scipy`, `matplotlib`, `seaborn`, `statsmodels`, and `scikit-learn` libraries.

Install the required libraries with:

```
pip install numpy pandas scipy matplotlib seaborn statsmodels scikit-learn
```

Statistical analysis is the process of examining data to understand it more clearly and extract useful insights. It helps identify patterns, relationships, and trends, thereby supporting decision-making and forecasting.

Phân tích thống kê là gì?

- Phân tích thống kê giúp rút ra thông tin có ý nghĩa từ dữ liệu bằng cách phát hiện các mẫu, xu hướng và mối quan hệ.
- Nó chuyển dữ liệu thô thành thông tin có thể hành động để hỗ trợ ra quyết định và dự đoán.



Statistical analysis overview

Quick Check - Prerequisites

Question 1. What is the main objective of statistical analysis?

- A. Only to store data
- B. To understand data and extract useful information
- C. Only to create charts
- D. To completely replace humans in decision-making

Question 2. True or false? Statistical analysis can help identify patterns, relationships, and trends in data.

Statistical Analysis Process

Statistical analysis is usually carried out through a structured process to ensure that results are accurate and meaningful. This process supports effective data collection, preparation, analysis, and communication.



Statistical analysis process

1. Collect Data

Data collection is the first step in statistical analysis. The data must have sufficient reliability and quality for the results to be meaningful.

Possible Data Sources

- ▶ Surveys.
- ▶ Observations.
- ▶ Experiments.
- ▶ Internal databases.
- ▶ Administrative data.
- ▶ APIs.
- ▶ Public datasets.
- ▶ Sensors.

Data Requirements

1. Collect Data

- ▶ Relevant to the research objective.
- ▶ Collected from trustworthy sources.
- ▶ Sufficient in size.
- ▶ Collected using a clear method.
- ▶ Designed to minimize bias and missingness.

Example

A company that wants to assess customer satisfaction may collect information through questionnaires, purchase histories, online feedback, and customer-service records.

Quick Check - 1. Collect Data

Question 1. Why should high-quality data be collected?

- A. To increase the number of columns
- B. To ensure reliable analytical results
- C. To eliminate the need for cleaning
- D. To replace interpretation

Question 2. Which can be used as a data source?

- A. Surveys
- B. APIs
- C. Databases
- D. All of the above

2. Organize the Data

After collection, data must be cleaned and organized before they can be analyzed correctly.

Main Tasks

- ▶ Use spreadsheets, databases, or programming tools.
- ▶ Handle missing values.
- ▶ Correct errors or inconsistent values.
- ▶ Remove duplicate records.
- ▶ Standardize formats.
- ▶ Check data types.
- ▶ Arrange the data in a suitable structure.

Possible Tools

- ▶ Microsoft Excel.
- ▶ Google Sheets.

2. Organize the Data

- ▶ SQL.
- ▶ Python with Pandas.
- ▶ Database software.

Python Example

```
import pandas as pd

df = pd.read_csv("data.csv")

print(df.head())
print(df.info())
print(df.isnull().sum())
```

Handle Missing Data

2. Organize the Data

```
df["age"] = df["age"].fillna(  
    df["age"].median()  
)
```

Remove Duplicates

```
df = df.drop_duplicates()
```

Quick Check - 2. Organize the Data

Question 1. Which activity belongs to data organization?

- A. Handling missing values
- B. Correcting inconsistent data
- C. Removing duplicates
- D. All of the above

Question 2. Which tool is suitable for handling tabular data in Python?

- A. Pandas
- B. Matplotlib
- C. Seaborn
- D. TensorFlow

Question 3. True or false? Collected data are always immediately ready for analysis.

3. Analyze the Data

At this stage, statistical techniques are applied to explore the data and extract useful insights.

Common Methods

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to understand the data and identify patterns, trends, and unusual observations.

Common techniques include:

- ▶ Descriptive statistics.
- ▶ Histograms.
- ▶ Box plots.
- ▶ Scatter plots.
- ▶ Correlation matrices.
- ▶ Outlier detection.

3. Analyze the Data

Confirmatory Data Analysis

Confirmatory Data Analysis (CDA) is used to test hypotheses or confirm conclusions formulated in advance.

Regression Analysis

Regression analysis studies the relationship between a dependent variable and one or more independent variables.

Examples:

- ▶ Predicting sales from advertising expenditure.
- ▶ Estimating house prices from area and location.
- ▶ Studying the effect of study time on examination scores.

Hypothesis Testing

Hypothesis testing evaluates whether an observed result is statistically significant or may have occurred by chance.

Quick Check - 3. Analyze the Data

Question 1. What is EDA mainly used for?

- A. Exploring data and identifying patterns
- B. Only presenting reports
- C. Only storing data
- D. Only creating optimization models

Question 2. CDA is commonly used to:

- A. Test hypotheses
- B. Delete data
- C. Create spreadsheets
- D. Collect data

Question 3. Regression analysis is used to:

- A. Study relationships among variables

Quick Check - 3. Analyze the Data

- B. Process text only
- C. Create pie charts only
- D. Remove every outlier

Question 4. True or false? Hypothesis testing helps determine whether a result is statistically significant.

4. Interpret and Present Results

After analysis, results must be explained and presented clearly so that others can understand and use them.

Presentation Formats

- ▶ Reports.
- ▶ Charts.
- ▶ Graphs.
- ▶ Dashboards.
- ▶ Presentations.
- ▶ Statistical tables.
- ▶ Executive summaries.

Presentation Principles

- ▶ State the analytical objective clearly.
- ▶ Explain the method used.

4. Interpret and Present Results

- ▶ Present the main findings.
- ▶ Discuss practical meaning.
- ▶ State limitations.
- ▶ Recommend next actions.
- ▶ Avoid overinterpreting statistical results.

Example

If an analysis shows that sales declined sharply in one region, a report should not provide only the percentage decline. It should explain:

- ▶ Which region was affected.
- ▶ How large the decline was.
- ▶ How long the trend lasted.
- ▶ Which factors may be related.
- ▶ Which actions the company should consider.

Quick Check - 4. Interpret and Present Results

Question 1. Which format can be used to present results?

- A. Reports
- B. Charts
- C. Dashboards
- D. All of the above

Question 2. Why should analytical limitations be stated?

- A. So readers understand the scope and reliability of the results
- B. To make the report longer
- C. To avoid presenting results
- D. To replace the data

Types of Statistical Analysis

There are six major types of statistical analysis:

- 1 Descriptive statistics.
- 2 Inferential statistics.
- 3 Exploratory Data Analysis.
- 4 Predictive modeling.
- 5 Prescriptive analysis.
- 6 Causal analysis.

Types of statistical analysis

Các loại phân tích thống kê



1) Phân tích khám phá



2) Thống kê mô tả



3) Thống kê suy luận



4) Phân tích dự đoán



5) Phân tích nhân quả



6) Phân tích khuyến nghị

Types of statistical analysis

1. Descriptive Statistics

Descriptive statistics summarize and organize data so that readers can quickly understand the main characteristics.

Common Techniques

- ▶ Measures of central tendency.
- ▶ Variance.
- ▶ Standard deviation.
- ▶ Histograms.
- ▶ Bar charts.
- ▶ Box plots.

Central Tendency

Common measures include:

- ▶ **Mean:** The sum of values divided by the number of observations.

1. Descriptive Statistics

- ▶ **Median:** The middle value after sorting.
- ▶ **Mode:** The most frequently occurring value.

Mean Formula

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Sample Variance

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Sample Standard Deviation

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

1. Descriptive Statistics

Python Example

```
print(df.describe())  
print(df["income"].mean())  
print(df["income"].median())  
print(df["income"].std())
```

Quick Check - 1. Descriptive Statistics

Question 1. Descriptive statistics are used to:

- A. Summarize data
- B. Prove causation
- C. Build deep-learning models only
- D. Forecast the future only

Question 2. Which measure describes data dispersion?

- A. Standard deviation
- B. Mode
- C. Median
- D. Variable name

Question 3. Which chart is commonly used to detect outliers?

- A. Box plot

Quick Check - 1. Descriptive Statistics

- B. Pie chart
- C. Line chart
- D. Map

2. Inferential Statistics

Inferential statistics use sample data to draw conclusions or make predictions about a larger population.

Common Techniques

- ▶ Hypothesis testing.
- ▶ Confidence intervals.
- ▶ Regression analysis.
- ▶ Analysis of variance.
- ▶ Parameter estimation.

Sample and Population

- ▶ **Population:** The complete group of interest.
- ▶ **Sample:** A subset selected from the population.

2. Inferential Statistics

Example

A university has 20,000 students. Instead of surveying all students, a researcher may select a sample of 1,000 students to estimate overall satisfaction.

Confidence Interval

A confidence interval provides a range of plausible values for a population parameter.

Quick Check - 2. Inferential Statistics

Question 1. Inferential statistics use sample data to:

- A. Draw conclusions about a population
- B. Describe the sample only
- C. Create charts only
- D. Remove data

Question 2. Which technique belongs to inferential statistics?

- A. Hypothesis testing
- B. Confidence intervals
- C. ANOVA
- D. All of the above

Question 3. True or false? A sample is the complete group of interest.

3. Exploratory Data Analysis

EDA focuses on understanding data before model building.

Objectives

- ▶ Understand data structure.
- ▶ Detect patterns.
- ▶ Examine relationships.
- ▶ Detect missing data.
- ▶ Identify outliers.
- ▶ Check initial assumptions.

Common Techniques

- ▶ Scatter plots.

3. Exploratory Data Analysis

- ▶ Correlation analysis.
- ▶ Distribution analysis.
- ▶ Outlier detection.

Python Example

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.histplot(
    df["income"],
    kde=True
)

plt.title("Income Distribution")
plt.show()
```

Quick Check - 3. Exploratory Data Analysis

Question 1. When is EDA commonly performed?

- A. Before model building
- B. Only after model deployment
- C. After deleting the data
- D. It is unrelated to modeling

Question 2. Which technique belongs to EDA?

- A. Scatter plots
- B. Correlation analysis
- C. Outlier detection
- D. All of the above

4. Predictive Modeling

Predictive models use historical data to forecast future outcomes or trends.

Common Methods

- ▶ Linear regression.
- ▶ Decision trees.
- ▶ Neural networks.
- ▶ Support Vector Machines.
- ▶ Random Forest.
- ▶ Time-series models.

Applications

- ▶ Sales forecasting.

4. Predictive Modeling

- ▶ Demand forecasting.
- ▶ Credit-risk prediction.
- ▶ Churn prediction.
- ▶ Price forecasting.

Python Example

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(
    X_train,
    y_train
)
```

4. Predictive Modeling

```
y_pred = model.predict(  
    X_test  
)
```

Quick Check - 4. Predictive Modeling

Question 1. Predictive models use:

- A. Historical data
- B. Text data only
- C. Missing data only
- D. No data

Question 2. Which method can be used for prediction?

- A. Linear regression
- B. Decision trees
- C. Neural networks
- D. All of the above

5. Prescriptive Analysis

Prescriptive analysis recommends the best actions based on data. It goes beyond prediction by proposing solutions that can help achieve desired outcomes.

Common Methods

- ▶ Optimization techniques.
- ▶ Simulation models.
- ▶ Decision theory.
- ▶ Scenario analysis.
- ▶ Multi-objective optimization.

Applications

- ▶ Selecting an optimal price.

5. Prescriptive Analysis

- ▶ Designing a production schedule.
- ▶ Optimizing delivery routes.
- ▶ Allocating an advertising budget.
- ▶ Selecting an investment portfolio.

Quick Check - 5. Prescriptive Analysis

Question 1. Which question does prescriptive analysis answer?

- A. What happened?
- B. Why did it happen?
- C. What may happen?
- D. What should we do?

Question 2. Which techniques are commonly used in prescriptive analysis?

- A. Optimization
- B. Simulation
- C. Decision theory
- D. All of the above

6. Causal Analysis

Causal analysis evaluates whether one variable causes a change in another.

Common Methods

- ▶ Randomized experiments.
- ▶ Regression models.
- ▶ Propensity-score matching.
- ▶ A/B testing.
- ▶ Instrumental variables.
- ▶ Difference-in-differences.

6. Causal Analysis

Example

A company wants to determine whether a promotion truly increases sales or merely coincides with a period of naturally rising demand.

Correlation and Causation

Correlation indicates that two variables are associated. It is not sufficient to establish causation.

For example:

- ▶ Ice-cream sales and the number of fires may both rise during summer.
- ▶ This does not mean that selling ice cream causes fires.
- ▶ High temperature may affect both variables.

Quick Check - 6. Causal Analysis

Question 1. The objective of causal analysis is to:

- A. Identify cause-and-effect relationships
- B. Summarize data only
- C. Calculate means only
- D. Create dashboards only

Question 2. Which method may be used in causal analysis?

- A. Randomized experiments
- B. Propensity-score matching
- C. Regression models
- D. All of the above

Question 3. True or false? A strong correlation always proves causation.

Python provides many libraries for data analysis and modeling.

Common Libraries

- ▶ **NumPy:** Numerical computing.
- ▶ **Pandas:** Data processing.
- ▶ **SciPy:** Statistical methods.
- ▶ **Scikit-learn:** Machine learning.
- ▶ **Matplotlib:** Visualization.
- ▶ **Seaborn:** Statistical visualization.
- ▶ **Statsmodels:** Statistical modeling.

SPSS is widely used in social-science research.

Strengths

- ▶ User-friendly interface.
- ▶ Supports many statistical tests.
- ▶ Suitable for users with limited programming experience.

Microsoft Excel

Excel is suitable for basic statistical calculations and simple visualizations.

Strengths

- ▶ Easy to use.
- ▶ Suitable for small datasets.
- ▶ Supports PivotTables.
- ▶ Provides basic statistical functions.

Comparison

Tool	Main purpose	Key feature
Python	Data analysis and modeling	Flexible and well integrated with machine learning
SPSS	Social-science research	User-friendly graphical interface
Excel	Basic analysis	Widely available and accessible

Quick Check - Statistical Analysis Tools

Question 1. Which Python libraries are commonly used for statistical methods?

- A. SciPy
- B. Pandas
- C. Statsmodels
- D. Both A and C

Question 2. Which tool is widely used in social-science research?

- A. SPSS
- B. Matplotlib
- C. NumPy
- D. Git

The Importance of Statistical Analysis

Statistical analysis plays an important role in decision-making because it helps convert data into evidence, uncertainty estimates, and performance measures.

- ▶ **Data-driven decision-making:** replaces guesswork with decisions based on data and evidence.
- ▶ **Evaluating uncertainty:** probability models help evaluate uncertainty and risk.
- ▶ **Testing ideas:** supports hypothesis testing and the confirmation of findings.
- ▶ **Evaluating performance:** helps measure KPIs, productivity, business growth, product quality, campaign effectiveness, and satisfaction levels.

Quick Check - The Importance of Statistical Analysis

Question 1. Statistical analysis helps replace:

- A. Guesswork with data-driven decisions
- B. Data with intuition
- C. Reports with speech
- D. Processes with randomness

Question 2. Probability models support:

- A. Evaluating uncertainty
- B. Eliminating all risk
- C. Removing data
- D. Creating charts only

Statistical analysis supports:

- ▶ Market research.
- ▶ Quality control.
- ▶ Customer analysis.
- ▶ Financial decision-making.
- ▶ Business-performance measurement.

Healthcare and Public Health

Statistical analysis is used in:

- ▶ Medical research.
- ▶ Public-health analysis.
- ▶ Drug-safety evaluation.
- ▶ Disease surveillance.
- ▶ Treatment-effectiveness assessment.

Statistical analysis can help:

- ▶ Evaluate learning outcomes.
- ▶ Improve teaching methods.
- ▶ Develop educational policy.
- ▶ Identify students who need support.

Statistical analysis supports:

- ▶ Human-behavior research.
- ▶ Social-trend analysis.
- ▶ Population studies.
- ▶ Public-policy evaluation.

Statistical analysis is used to:

- ▶ Analyze climate.
- ▶ Monitor pollution.
- ▶ Evaluate resources.
- ▶ Support conservation.
- ▶ Forecast environmental risk.

Quick Check - Environment

Question 1. Statistical analysis can be applied in:

- A. Healthcare
- B. Education
- C. The environment
- D. All of the above

Question 2. In education, statistical analysis can help:

- A. Evaluate learning outcomes
- B. Improve teaching methods
- C. Support policy development
- D. All of the above

Question 3. Case. A hospital wants to evaluate the effectiveness of a new treatment. How can statistical analysis support this task?

Content Summary

Topic	Main objective
Data collection	Obtain quality data from appropriate sources
Data organization	Clean and structure the data
Data analysis	Apply statistical techniques
Interpretation and presentation	Communicate results clearly
Descriptive statistics	Summarize data characteristics
Inferential statistics	Generalize from a sample to a population
EDA	Explore patterns and data problems
Predictive modeling	Forecast future outcomes
Prescriptive analysis	Recommend actions
Causal analysis	Identify cause-and-effect relationships